

Phase 1: Offline Runtime Demo

Short Description

The Offline Runtime Demo is the foundation layer of the SovAI Air-Gap AI Platform. It demonstrates how an AI application can be prepared while internet access is available, packaged into approved local artifacts, and then executed without runtime internet connectivity.

This phase does not focus on advanced AI generation. Instead, it proves the most important air-gap principle: once the runtime starts, it should depend only on local artifacts, local documents, approved tools, and internal logs.

The demo includes local document retrieval, a lightweight ML classifier, controlled agent tools, and audit logging. It is designed as a laptop-friendly reference implementation for sovereign AI and disconnected enterprise environments.

What This Phase Proves

- The application can run without internet after preparation.
- Required dependencies are packaged before going offline.
- Docker images are built and saved as local artifacts.
- Runtime services use local documents only.
- Agent tools are restricted to an approved list.
- Unapproved tools such as web search are blocked.
- All important actions are written to an audit log.

Repository

```
git clone <your-github-repo-url>
cd sovai-airgap-ai-starter
```

Prerequisites

- Docker Desktop installed
- Git Bash, WSL, or Linux/macOS terminal
- Internet access during preparation only

Setup: Connected Preparation Phase

Run this while internet is available:

```
chmod +x scripts/*.sh
./scripts/prepare-online.sh
```

This script performs the connected preparation step:

1. Builds the Docker application image
2. Installs required Python packages into the image
3. Saves the image as a local artifact
4. Downloads Python wheels into the local wheelhouse
5. Creates checksums
6. Creates an artifact bundle

Generated artifacts include:

```
artifact-hub/images/sovai-airgap-app.tar
artifact-hub/wheelhouse/
artifact-hub/checksums.sha256
sovai-airgap-artifact-bundle.tar.gz
```

Setup: Offline Runtime Phase

Disconnect internet first:

Turn off Wi-Fi, unplug network, or disable VPN.

Then run:

```
./scripts/bootstrap-offline.sh
```

This script:

1. Verifies host internet is disconnected
2. Loads the saved Docker image from artifact-hub/images
3. Starts the application using Docker Compose
4. Runs smoke tests

Open the application:

```
http://127.0.0.1:8080
```

Configuration

Main configuration file:

```
docker-compose.yml
```

Important environment variables:

```
SOVAI_OFFLINE_MODE: "true"  
APPROVED_TOOLS: "calculator,document_search,ticket_classifier"  
DOCUMENTS_DIR: "/app/data/documents"  
AUDIT_LOG_PATH: "/app/data/audit/audit-log.jsonl"
```

Documents are stored in:

```
data/documents/
```

Audit logs are written to:

```
data/audit/audit-log.jsonl
```

Approved tools are listed in:

```
artifact-hub/tools/approved-tools.yaml
```

Test Commands

Check health:

```
curl http://127.0.0.1:8080/health
```

Check status:

```
curl http://127.0.0.1:8080/status
```

Ask a private document question:

```
curl -s http://127.0.0.1:8080/ask
-H "Content-Type: application/json"
-d '{"question":"Who can access confidential finance reports?","top_k":3}'
```

Run local ML classifier:

```
curl -s http://127.0.0.1:8080/ml/classify
-H "Content-Type: application/json"
-d '{"text":"User cannot access VPN and needs MFA reset"}'
```

Run approved agent tool:

```
curl -s http://127.0.0.1:8080/agent/run
-H "Content-Type: application/json"
-d '{"tool":"calculator","input":"(12 * 7) + 5}"'
```

Test blocked tool:

```
curl -s http://127.0.0.1:8080/agent/run
-H "Content-Type: application/json"
-d '{"tool":"web_search","input":"latest AI news}"'
```

Expected result:

```
{
  "allowed": false
}
```

Troubleshooting

Problem: scikit-learn build failed on Windows

Earlier version used `scikit-learn`, `numpy`, and `scipy`. On Windows, pip sometimes downloaded the source package and tried to compile it, causing errors such as:

```
ERROR: Unknown compiler(s)
metadata-generation-failed
```

Fix:

The current version removes compiled ML dependencies and uses pure-Python retrieval and classification.

Use the patched version that only requires:

```
fastapi
uvicorn
pydantic
jinja2
```

Problem: Container runs but browser cannot access the app

Check:

```
docker compose ps
docker port sovai-airgap-app
```

Expected:

```
8080/tcp -> 0.0.0.0:8080
```

If no port is shown, recreate the container:

```
docker compose down --remove-orphans
docker rm -f sovai-airgap-app 2>/dev/null || true
docker compose up -d --no-build --force-recreate
```

Problem: localhost does not work

Use:

```
http://127.0.0.1:8080
```

instead of:

```
http://localhost:8080
```

Problem: Docker network blocks browser access

For the laptop demo, use:

```
networks:
  sovai_internal:
    driver: bridge
```

Avoid this for the first demo:

```
internal: true
```

because it can make local browser testing harder.

Problem: Port 8080 already used

Change the port mapping:

```
ports:
  - "8090:8080"
```

Then recreate:

```
docker compose down
docker compose up -d --no-build --force-recreate
```

Open:

```
http://127.0.0.1:8090
```

Positioning

This phase is not a full enterprise TL4 platform. It is a laptop reference implementation that demonstrates the offline runtime pattern:

```
Prepare while connected.
Run while disconnected.
Consume only local artifacts.
Log every important action.
```